

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - 2027	B.Tech.	Indian Institute of Technology Kanpur	9.0/10.0
2023	CBSE(XII)	Bal Bharati Public School, Dwarka	97.0%
2021	CBSE(X)	Bal Bharati Public School, Dwarka	96.4%

Scholastic Achievements

- Received the IIT Kanpur **Academic Excellence Award** in recognition of outstanding academic performance
 - Granted **branch change** from **Mechanical Engineering to CSE** based on excellent academic performance in first year
 - Recognised as **school topper** for outstanding performance in **Physics** in AISSCE examination securing **98%** marks
- '24

'24

'23

Internship Experience

Developer Intern | Samsung Research Institute, Bangalore

(June'26-Jul'26)

Objective	- Develop a skill for end-to-end API testing pipeline given an OpenAPI specification for the native coding assistant - Build a remediation skill suggesting fixes for functional, non-functional, and security issues found during scans
Approach	- Designed an 8-phase skill using Chain of Thought, ReAct and other state-of-the-art prompting techniques - Generated functional and non-functional tests covering numerous integration, load, stress, volume, endurance tests - Developed support for Karate, Postman, RestAssured, SuperTest, k6, JMeter, Gatling, and Locust frameworks - Containerized mock and parent server and test execution suites for accurate performance estimation using Docker - Assessed the codebase for functional, non-functional, security issues and proposed remedies for better performance
Result	Reduced human intervention by 3x, from 8-10 hours to just 2-3 hours and a minimal 5-10 clicks per testing. - Delivered accurate, actionable fixes using the remediation skill at line-level granularity for identified issues based.

Research Projects

Leveraging M1 reward signals for adaptive BCI | Prof. Arjun Ramakrishnan | SURGE

(May'25-Jul'25)

Objective	Harness motor cortical (M1) reward signals to enhance real time adaptability of BCIs making them more realistic
Approach	- Epoched EEG data using event markers, applied baseline correction and normalization for faster convergence - Trained LSTM, EEGNET, and DSCNN+LSTM models using categorical cross-entropy and Adam optimizer - Developing state-of-the-art deep learning model quantifying M1 reward signals as RL agent's reward function
Result	Achieved a maximum accuracy of 45% using EEGNET, closely approaching performance of S.O.T.A. models

Key Projects

VitalEye - Contactless Vital Sign Monitoring System | Prof. Priyanka Bagade | CS776

(Jan'26-Apr'26)

- Designed a modular pipeline to estimate heart rate and respiratory rate from RGB facial video for improved monitoring of vitals
 - Integrated **YOLOv8** ROI detection, spatiotemporal modeling, and FFT-based analysis improving HR estimation by **3-5 BPM**
 - Implemented **PhysFormerX**, a 3D residual backbone with **Squeeze-and-Excitation** attention and a temporal refinement head
 - Trained using biologically significant composite **Pearson correlation**, **frequency cross-entropy**, and **Smooth-L1** loss
 - Benchmarked 8+ architectures (**HybridRPPG**, **PhysFormerX**) on UBFC-rPPG dataset, with a test **MAE of 6.28 BPM**
 - Deployed the system integrating OpenVINO CPU acceleration for a **5x-10x inference speedup** and real-time inference

normaldb - 3NF Relational Schema Synthesizer | Prof. Arnab Bhattacharya | CS315

(Jan'26-Apr'26)

- Implemented **Bernstein's** synthesis algorithm to generate relational schemas in **3NF** form given a set of functional dependencies
 - Integrated the **linear-time Beeri-Bernstein** algorithm to efficiently check functional dependency membership in a closure set
 - Built a **Flask-based GUI** accepting user-defined functional dependencies and rendering the synthesized **Codd's 3NF** schema
 - Maintained code quality using isort and black formatting standards, validated through continuous CodeFactor analysis

Single-Trial EEG Decoding of Auditory Prediction Errors | Prof. Arjun Ramakrishnan | BSE662

(Jan'26-April'26)

- Trained **EEGNet CNN** and **LSTM-hybrid** models for binary and multi-class deviant-versus-standard trial classification tasks
 - Remodelled the task as regression, to predict cosine similarity to subject-specific MMN templates to get **Pearson r=0.78**
 - Built **T-GCN** and **ST-GAT GNNs** modeling inter-electrode correlations and pathways, acheiving **61%** classification accuracy
 - Validated neural representations using **permutation testing** and **leave-one-subject-out cross-validation** across 34 subjects

Mental Workload Localisation using N-Back Task | Prof. Arjun Ramakrishnan

(May'25-July'25)

- Conducting **EEG analysis** to identify neural signatures across cognitive load levels, from low-demand to high-intensity tasks
 - Preprocessed data by manually removing bad channels, applying **Independent Component Analysis** for eye artifact removal
 - Produced topomaps and average band power plots for delta through gamma bands for **comparative frequency-band analysis**
 - Plotted spectrograms** for different channels estimating the effect of breaks and different mental workloads on EEG signals

IITK mini MIPS - A mini processor 🧠 | Prof. Debapriya Basu Roy | CS220 (Mar'25-Apr'25)

- Designed instruction and data memory, and instruction decoder on **FPGA** with **Verilog HDL**, optimizing system performance
- Implemented a **32-bit ALU** in Verilog, supporting arithmetic, logical, shift, and jump operations as per the **MIPS-32 ISA**
- Developed a structured data path and control logic, incorporating a **finite state machine** for efficient instruction execution

Brokeo - Finance Tracking Application 🧠 | Prof. Indranil Saha | CS253 (Jan'25-Apr'25)

- Developed a **finance tracking application** to efficiently **monitor** and **manage** personal transactions, dues, and expenditure
- Drafted **SRS** and **Design documents** detailing app requirements, design specifications, and system implementation constraints
- Authored **Implementation**, **Beta Testing**, and **User Manual** documents including bug reports and application functionality
- Designed **Figma** prototypes for **10** app screens incorporating comprehensive flows, responsive layouts, and interactive elements
- Built **Flutter** front-end for analytics and merchant management pages, leveraging rich widgets to visualize transaction trends

OpenViBE for BCI 🧠 | Prof. Nikunj Arunkumar Bhagat (Sep'24-May'25)

- Developed a **real-time** P300 paradigm based Brain-Computer Interface in **OpenViBE**, processing **100+ EEG samples/sec**
- Leveraged **MATLAB** and Lab Streaming Layer (LSL) to stream real time EEG data into OpenViBE for a P300 Speller paradigm.
- Applied custom **Lua scripts** within OpenViBE to enhance control and efficiency of the target letter generation processes
- Explored **OpenViBE-Arduino communication** interface to stream real time BCI outputs to the **microcontroller**
- Integrating **Python** into OpenViBE's pipeline to enable flexible, customizable EEG **signal processing** and **predictions**

DeepFake Detection 🧠 | Brain and Cognitive Science Club, IITK (Sep'24-Oct'24)

- Developed a deepfake detection system implementing **DeepFakeNet** using **PyTorch** to enhance integrity of multimedia content
- Studied existing literature on usage of **CNNs**, **MesoNet**, **Residual Networks (ResNets)**, **ResNeXts** in deepfake detection
- Compiled a comprehensive dataset of **over 10,000** natural and manipulated **face images** ensuring balanced class distribution
- Implemented **flips**, **rotation**, and **scaling** augmentations to improve model generalization, boosting validation accuracy to **93%**

Skills

Programming Languages: Python, C++, JavaScript, Bash, Verilog, Haskell | **Other Tools:** Git, Github, SQL, MATLAB, L^AT_EX

Libraries: Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn, PyTorch, OpenCV, Tensorflow, Keras, MNE-Python, torchvision

Positions of Responsibility

Core Team Member | Institute Counselling Service, IITK (Apr'25-Apr'26)

- Liaising with 9 counsellors to provide emotional & academic help to a campus community of 4500+ undergraduate students
- Managing 300+ Student Guides and Academic Mentors in a 7 day orientation for 1200+ freshers as a part of a 22 member team
- Organizing 4-day training workshops for 300+ Student Guides and Academic Mentors to train them for supporting freshers

Secretary | Brain and Cognitive Science Club (BCS), IITK (Jul'24-April'25)

- Received **Best Mentor** award for exceptional guidance and support during winter projects organised by the BCS club, IITK
- Conducted a workshop on **Generative Adversarial Networks** in Neural Net November, the flagship event of BCS club
- Mentored a group of **14** mentees on foundational topics in ML and Convolutional Neural Networks for Image Classification

Academic Mentor | Institute Counselling Service, IITK (Aug'24-April'25)

- Applied training strategies to teach **TA111-Technical Arts**, to over **20 freshmen** and delivered wing level sessions

Relevant Courses

*:Excellent Performance | **:Outstanding Performance | † : Ongoing

Data Structures and Algorithms	Linear Algebra	Computer Organisation
Fundamentals of Computing*	Discrete Mathematics	Ordinary Differential Equations*
Data Science and Machine Learning*	Software Development	Probability
Computer Vision and Deep Learning*	Computer Networks*	Principles of Database Systems
Probabilistic Machine Learning	Decision Making and the Brain*	Digital Electronics†
Principles of Programming Languages†	Foundations of Modern AI†	Principles of Communications†

Co-Scholastic achievements

- Co-founded the **school programming club** upskilling over **20 students** in Python programming and app development '21
- Awarded **Best Delegate** at UNNATI's Satta Power-of-Words MUN, serving on the **World Economic Forum** committee '20
- Bagged **II position** in **ATL Hackathon** by **GoI** for a **Python** App helping class 10 students manage their study resources '20
- Bagged **III position** in culinary event in IITK inter-hall cultural competition - Galaxy as a part of General Championship '24